on the motor shaft is 1m while the centre distance between pulleys is 1.75 m. The belt velocity is 1600 m/min. Determine the number of V-belts required to transmit the power if each belt has a cross sectional area of 375 mm^2, mas per metre length is 0.375 kg and an allowable stress of 2.5 MPa. The groove angle of the pulley is 35^0. The coefficient of friction between the belt and the pulley is 0.25. (14)

OR

X a) Derive an expression for the ratio of maximum and minimum tensions in the brake straps of a band and block brake. (8)

b) A cone clutch has a semi - cone angle 20^0, inner radius 100 mm and outer radius 150 mm. The material combination has a coefficient of friction of 0.25 and the allowable normal pressure is 20 N/cm^2. Find the necessary axial load and power that can be transmitted at 950 rpm. (12)

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Code No. BTS 012(E)

B.Tech. Degree V Semester Examination in Mechanical Engineering (CAD/CAM), April 2000

ME 501 MECHANICS OF MACHINERY

Time: 3 Hours *Max. Marks: 100*

I a) What are the differences between whitworth quick return motion and crank and slotted lever quick return motion ? Explain with sketches. (10)

b) In a crank and slotted lever quick return motion mechanism, the distance between the fixed centres is 240 mm and the length of the driving crank is 120mm. Find the inclination of the slotted bar with the vertical in the extreme position and time ratio of cutting stroke to the return stroke. If the length of the slotted bar is 450 mm, find the length of the stroke if the line of stroke passes through the extreme positions of the free end of the lever. (10)

OR

II a) State and explain the Kennedy's theorem of three centers. (5)

b) The crank of a slider crank mechanism rotates clockwise at a constant speed of 300 rpm. The crank is 15 cm and the connecting rod is 60 cm long. Determine (a) linear velocity and acceleration of the mid point of the connecting rod (b) angular velocity and angular acceleration of the connecting rod, at a crank angle of 45^0 from IDC position. (15)

III a) What do you understand by coupler curves ? Describe the method of obtaining the co-ordinates of a coupler point in a slider crank mechanism. (8)

b) Synthesize a four bar mechanism to generate the function y = log x, where x varies between 1 and 10. Use three accuracy points with chebychev's spacing. Assume $\theta_s = 45^0$; $\theta_F = 105^0$; $\phi_s = 135^\circ$ and $\phi_F = 225^\circ$. Take the length of the smallest link equal to 50mm. (12)

(P.T.O)

OR

IV The three conditions to be satisfied by a four bar linkage are

$$\theta_2 = 60^\circ \qquad \theta_4 = 90^\circ$$

$$\omega_2 = \frac{d\theta_2}{dt} = 3 rad/s \qquad \omega_4 = \frac{d\theta_4}{dt} = 2 \ rad/s$$

$$\alpha_2 = \frac{d^2\theta_2}{dt^2} = -1 \ rad/s^2 \qquad \alpha_4 = \frac{d^2\theta_4}{dt^2} = 0$$

Determine the link length ratios. Sketch the mechanism taking the length of the shortest link as 50 mm. (20)

V a) Explain with neat sketch, any one type of intermittent motion mechanism. (6)

b) Design a cam for operating the exhaust valve of an oil engine. It is required to give equal uniform acceleration and retardation during opening and closing of the valve each of which corresponds to 60^0 of cam rotation. The valve must remain in the fully open position for 20^0 of cam rotation. The lift of the valve is 37.5 mm and the least radius of the cam is 50 mm. The follower is provided with a roller of 50 mm diameter and its line of stroke passes through the axis of the cam. (14)

OR

VI A cam has straight working faces which are tangential to a base circle of diameter 90mm. The follower is a roller of diameter 40mm and the centre of the roller moves along a straight line passing through the centre line of the cam shaft. The angle between the tangential faces of the cam is 90^0 and the faces are joined by a nose circle of 10mm radius. The speed of rotation of the cam is 120 revolutions per min. Find the acceleration of the roller centre when (a) during the lift, the roller is just about to leave the straight flank and (b) the roller is at the outer end of its lift. (20)

Contd...3

VII a) Derive an expression for the length of path of contact for two meshing spur gears having involute profile. (8)

b) A pinion having 20 teeth of involute form, 20^0 pressure angle and 6 mm module drives a gear having 40 teeth. If addendum is equal to module find (a) addendum and pitch circle radii of the two gears, (b) length of path of approach and contact and (c) length of arc of contact. (12)

OR

VIII Figure 1 shows a gear train used in ships. It is seen that the shaft 'p' is stationary if the propellers run at the same speed and drive the gears C and D in the same direction through equal gears A and B. The number of teeth on G and F are 24 and 50 respectively. Find the ratio of number of teeth on C and D. If the speed of C is 100 rpm and B runs 5% faster than A, find the speed of P. H is a planet gear mounted on the arm E.
E, D and C are concentric. (20)

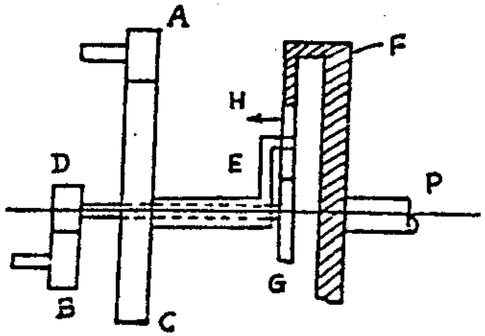

Fig 1

IX a) Obtain the condition for the maximum power transmitted by a belt from one pulley to another. (6)

b) A compressor, requiring 96 kw is to run at about 750 rpm. The drive is by V-belts from an electric motor running at 250 rpm. The diameter of the pulley

Contd...4